Evaluation runs: **replog** (replicated-log-aneris)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **asyn-fault**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	6	36	conforms	0.0	4	1 h, 250 GB	3	9	brain03, 2026-11-02
3	20	400	conforms	0.1	4	1 h, 250 GB	6	41	brain03, 2026-11-02
4	70	4900	conforms	0.2	4	1 h, 250 GB	11	230	brain03, 2026-11-02
5	252	63504	conforms	0.8	4	1 h, 250 GB	24	1416	brain03, 2026-11-02
6	924	853776	conforms	8.4	10	1 h, 250 GB	55	12007	brain03, 2026-11-02
7	3432	11778624	conforms	120.8	78	1 h, 250 GB	264	278608	brain03, 2026-11-02
8	12870		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	12	30	conforms	0.0	4	1 h, 250 GB	3	14	brain03, 2026-11-02
3	120	280	conforms	0.1	4	1 h, 250 GB	7	182	brain03, 2026-11-02
4	1680	2940	conforms	0.5	4	1 h, 250 GB	22	8776	brain03, 2026-11-02
5	30240	33264	conforms	12.5	4	1 h, 250 GB	85	651755	brain03, 2026-11-02
6	665280	396396	conforms	404.9	6	1 h, 250 GB	287	34200952	brain03, 2026-11-02
7	17297280		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	6	30	conforms	0.0	4	1 h, 250 GB	3	9	brain03, 2026-11-02
3	20	280	conforms	0.1	4	1 h, 250 GB	6	41	brain03, 2026-11-02
4	70	2940	conforms	0.2	4	1 h, 250 GB	13	275	brain03, 2026-11-02
5	252	33264	conforms	0.8	4	1 h, 250 GB	26	1629	brain03, 2026-11-02
6	924	396396	conforms	4.2	6	1 h, 250 GB	85	18571	brain03, 2026-11-02
7	3432	4907760	conforms	47.0	18	1 h, 250 GB	148	99875	brain03, 2026-11-02
8	12870	62573940	conforms	774.9	286	1 h, 250 GB	323	771832	brain03, 2026-11-02
9	48620		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	12	144	12	violation	0.1	4	1 h, 250 GB	16	161	brain03, 2026-11-02
3	120	14400	2880	violation	6.0	4	1 h, 250 GB	2889	345834	brain03, 2026-11-02
4	1680			(1 h)			1 h, 250 GB			brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 7 correct bag, 8 correct fifo, 8 fifo asyn-fault, 8 fifo correct, 8 asyn-fault fifo, 9 fifo correct, 9 fifo asyn-fault, 10 fifo correct, 10 fifo asyn-fault.

skipped: a smaller size had reached its limit: 5 asyn-fault bag, 6 asyn-fault bag, 7 asyn-fault bag, 8 correct bag, 8 asyn-fault bag, 9 asyn-fault fifo, 9 asyn-fault bag, 9 correct bag, 10 asyn-fault fifo, 10 correct fifo, 10 correct bag.

stateful

stateful: pair **asyn-fault**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	6	36	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3	20	400	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
4	70	4900	conforms	0.2	4	1 h, 250 GB	brain02, 2026-11-02
5	252	63504	conforms	0.7	4	1 h, 250 GB	brain02, 2026-11-02
6	924	853776	conforms	7.4	10	1 h, 250 GB	brain02, 2026-11-02
7	3432	11778624	conforms	124.8	64	48 h, 250 GB	brain03, 2026-11-06
8	12870	165636900	conforms	2557.7	1068	48 h, 250 GB	brain05, 2026-11-02
9	48620	2363904400	conforms	41296.0	14812	48 h, 250 GB	brain05, 2026-11-02
10			(48 h)			48 h, 250 GB	brain01, 2026-11-05
11			(48 h)			48 h, 250 GB	brain04, 2026-11-03
14			(48 h)			48 h, 200 GB	volta11, 2026-11-05

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	12	30	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3	120	280	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4	1680	2940	conforms	0.3	4	1 h, 250 GB	brain02, 2026-11-02
5	30240	33264	conforms	1.1	4	1 h, 250 GB	brain02, 2026-11-02
6	665280	396396	conforms	9.4	6	1 h, 250 GB	brain02, 2026-11-02
7	17297280	4907760	conforms	200.0	58	48 h, 250 GB	brain03, 2026-11-06
8			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	6	30	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3	20	280	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4	70	2940	conforms	0.2	4	1 h, 250 GB	brain02, 2026-11-02
5	252	33264	conforms	0.8	4	1 h, 250 GB	brain02, 2026-11-02
6	924	396396	conforms	4.3	4	1 h, 250 GB	brain02, 2026-11-02
7	3432	4907760	conforms	50.3	34	48 h, 250 GB	brain03, 2026-11-06
8	12870	62573940	conforms	894.4	290	48 h, 250 GB	brain05, 2026-11-02
9	48620	816621520	conforms	14594.0	3194	48 h, 250 GB	brain05, 2026-11-02
10			(48 h)			48 h, 250 GB	brain04, 2026-11-02

stateful: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	12	144	12	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
3	120	14400	2880	violation	0.5	4	1 h, 250 GB	brain02, 2026-11-01
4	1680	2822400	912240	violation	19.0	10	1 h, 250 GB	brain02, 2026-11-01

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
5				(1 h)			1 h, 250 GB	brain02, 2026-11-01

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 5 asyn-fault bag, 9 asyn-fault fifo, 11 asyn-fault fifo, 11 correct fifo, 12 fifo correct, 12 correct fifo, 12 asyn-fault fifo, 13 correct fifo, 13 asyn-fault fifo, 14 correct fifo, 15 correct fifo, 15 asyn-fault fifo, 16 correct fifo, 16 asyn-fault fifo.

skipped: a smaller size had reached its limit: 6 asyn-fault bag, 7 asyn-fault bag, 8 asyn-fault bag, 9 asyn-fault bag, 9 correct bag, 10 asyn-fault fifo, 10 correct fifo, 10 correct bag.

left out by the table rules (ok): 7 asyn-fault fifo, 7 correct fifo, 7 correct bag, 7 correct fifo.x3.t16.c48h.r7, 7 correct bag.x3.t16.c48h.r7, 7 asyn-fault fifo.x3.t16.c48h.r7, 8 correct fifo, 8 asyn-fault fifo.

left out by the table rules (timeout): 9 asyn-fault fifo, 9 correct fifo.

left out by the table rules (invalid: foreign load): 11 fifo correct, 12 fifo asyn-fault.

offline

offline: pair **asyn-fault**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	36	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	20	400	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
4	70	4900	conforms	0.3	8	10 min, 16 GB	0.0	0.3	0.0	in-memory	brain05, 2026-11-02
5	252	63504	conforms	1.1	8	10 min, 16 GB	0.1	1.1	0.0	in-memory	brain05, 2026-11-02
6	924	853776	conforms	8.6	20	10 min, 16 GB	0.1	8.5	0.1	in-memory	brain05, 2026-11-02
7	3432	11778624	conforms	145.2	108	48 h, 250 GB	0.3	144.8	0.4	merge	brain03, 2026-11-06
8	12870	165636900	conforms	2958.3	1144	1 h, 250 GB	0.8	2956.4	1.9	merge	brain05, 2026-11-03

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	30	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3	120	280	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	1680	2940	conforms	0.4	8	10 min, 16 GB	0.1	0.4	0.0	in-memory	brain05, 2026-11-02
5	30240	33264	conforms	1.2	32	10 min, 16 GB	0.7	1.1	0.2	in-memory	brain05, 2026-11-02
6	665280	396396	conforms	9.4	74	1 h, 250 GB	7.8	5.2	1.6	merge	brain05, 2026-11-03
7	17297280	4907760	conforms	192.3	642	48 h, 250 GB	183.7	60.5	8.6	merge	brain03, 2026-11-06

offline: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	30	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	20	280	conforms	0.2	8	10 min, 16 GB	0.0	0.2	0.0	in-memory	brain05, 2026-11-02
4	70	2940	conforms	0.3	8	10 min, 16 GB	0.0	0.3	0.0	in-memory	brain05, 2026-11-02
5	252	33264	conforms	1.1	28	10 min, 16 GB	0.1	1.0	0.1	in-memory	brain05, 2026-11-02
6	924	396396	conforms	7.3	72	1 h, 250 GB	0.1	5.7	1.6	merge	brain05, 2026-11-03
7	3432	4907760	conforms	69.2	559	48 h, 250 GB	0.3	60.8	8.3	merge	brain03, 2026-11-06
8	12870	62573940	conforms	1292.2	4588	1 h, 250 GB	0.8	1195.8	96.4	merge	brain05, 2026-11-03

offline: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	144	1	violation	0.1	8	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
3	120	14400	10	violation	0.6	8	1 h, 250 GB	0.1	0.5	0.1	merge	brain05, 2026-11-03
4	1680	2822400	74	violation	18.6	26	1 h, 250 GB	0.1	18.4	0.3	merge	brain05, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 5 asyn-fault bag, 8 fifo asyn-fault, 8 fifo correct, 8 asyn-fault fifo, 9 asyn-fault fifo, 9 correct fifo, 9 fifo correct, 9 fifo asyn-fault, 9 correct bag, 10 fifo asyn-fault.

skipped: a smaller size had reached its limit: 5 asyn-fault bag, 6 asyn-fault bag, 7 asyn-fault bag, 8 asyn-fault bag, 9 asyn-fault fifo, 9 correct fifo, 9 asyn-fault bag, 9 correct bag, 10 asyn-fault fifo, 10 correct fifo, 10 fifo correct, 10 correct bag.

left out by the table rules (ok): 7 asyn-fault fifo, 7 correct fifo, 7 correct bag, 7 correct bag.x3.m.c48h.r7, 7 correct fifo.x3.m.c48h.r7, 7 asyn-fault fifo.x3.m.c48h.r7.

refused (an input missing at run time): 8 correct bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	30	conforms	6.4	169	4 h, 250 GB	0.1	6.4	0.0	merge	brain05, 2026-11-03
3	120	280	conforms	9.2	166	10 min, 250 GB	0.1	9.1	0.1	merge	brain01, 2026-11-03
4	1680	2940	conforms	21.6	3071	10 min, 250 GB	0.1	21.6	0.1	merge	brain01, 2026-11-03

cross offline, Java implementation against the Rust specification: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	144	1	violation	4.7	163	10 min, 250 GB	0.1	4.7	0.0	merge	brain01, 2026-11-03
3	120	14400	10	violation	25.4	4412	10 min, 250 GB	0.1	25.1	0.3	merge	brain01, 2026-11-03

Not run (cross offline, Java implementation against the Rust specification). skipped: a smaller size had reached its limit: 4 bag asyn-fault, 10 fifo correct.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 8 fifo asyn-fault, 8 fifo correct, 9 fifo correct, 9 fifo asyn-fault, 10 fifo asyn-fault.

refused (an input missing at run time): 2 fifo correct, 2 fifo asyn-fault, 3 fifo asyn-fault, 3 fifo correct, 4 fifo asyn-fault, 4 fifo correct.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **asyn-fault**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	36	conforms	4.3	162	10 min, 250 GB	4.1	0.1	0.2	merge	brain01, 2026-11-03
3	20	400	conforms	8.1	155	10 min, 250 GB	8.0	0.1	0.1	merge	brain01, 2026-11-03
4	70	4900	conforms	8.2	151	10 min, 250 GB	8.2	0.3	0.0	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	30	conforms	6.5	152	4 h, 250 GB	6.5	0.0	0.0	merge	brain05, 2026-11-03
3	120	280	conforms	4.5	161	10 min, 250 GB	4.5	0.1	0.0	merge	brain01, 2026-11-03
4	1680	2940	conforms	8.4		10 min, 250 GB	8.4	0.4	0.1	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	6	30	conforms	6.1	160	4 h, 250 GB	6.1	0.1	0.0	merge	brain05, 2026-11-03
3	20	280	conforms	8.3	155	10 min, 250 GB	8.0	0.2	0.3	merge	brain01, 2026-11-03
4	70	2940	conforms	8.3	151	10 min, 250 GB	8.2	0.3	0.2	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	144	1	violation	3.9	153	10 min, 250 GB	3.9	0.1	0.0	merge	brain01, 2026-11-03
3	120	14400	10	violation	4.6	161	10 min, 250 GB	4.5	0.5	0.1	merge	brain01, 2026-11-03
4	1680	2822400	74	violation	26.3	271	48 h, 250 GB	6.1	26.3	0.0	merge	brain15, 2026-11-07

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 fifo asyn-fault, 5 fifo correct, 8 fifo asyn-fault, 8 fifo correct, 9 fifo correct, 9 fifo asyn-fault, 10 fifo asyn-fault.

skipped: a smaller size had reached its limit: 10 fifo correct.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
10	184756		(48 h)			48 h, 200 GB		volta10, 2026-11-05

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 2 bag asyn-fault, 2 fifo correct, 2 bag correct, 2 fifo asyn-fault, 3 bag correct, 3 bag asyn-fault, 3 fifo asyn-fault, 3 fifo correct, 4 fifo asyn-fault, 4 fifo correct, 4 bag correct, 8 fifo asyn-fault, 8 fifo correct, 9 fifo correct, 9 fifo asyn-fault, 10 fifo asyn-fault.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **asyn-fault**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	6	36	conforms	3.2	160	1 h, 250 GB		brain01, 2026-11-02
3	20	400	conforms	3.3	161	1 h, 250 GB		brain01, 2026-11-02
4	70	4900	conforms	3.8	162	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	12	30	conforms	3.1	160	1 h, 250 GB		brain01, 2026-11-02
3	120	280	conforms	3.6	167	1 h, 250 GB		brain01, 2026-11-02
4	1680	2940	conforms	7.7	583	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	6	30	conforms	3.2	160	1 h, 250 GB		brain01, 2026-11-02
3	20	280	conforms	3.3	161	1 h, 250 GB		brain01, 2026-11-02
4	70	2940	conforms	3.8	162	1 h, 250 GB		brain01, 2026-11-02
10	184756		oom (80 GB)			48 h, 80 GB		pinky02, 2026-11-05

cross online, Rust implementation (TraceForge) against the Java specification export: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	12	144	12	violation	3.1	160	1 h, 250 GB	2	brain01, 2026-11-02
3	120	14400	2880	violation	4.1	167	1 h, 250 GB	4	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 8 fifo asyn-fault, 8 fifo correct, 9 fifo correct, 9 fifo asyn-fault, 10 fifo asyn-fault.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **asyn-fault**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	144	144	conforms	5.2	163	10 min, 250 GB	4.7	0.1	0.6	merge	brain01, 2026-11-03
3	14400	14400	conforms	25.2	4412	10 min, 250 GB	25.1	0.5	0.1	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **asyn-fault**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	36	36	conforms	4.1	160	10 min, 250 GB	4.1	0.1	0.1	merge	brain01, 2026-11-03
3	400	400	conforms	9.6	160	10 min, 250 GB	9.6	0.1	0.0	merge	brain01, 2026-11-03
4	4900	4900	conforms	15.1	3371	10 min, 250 GB	14.9	0.3	0.2	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	30	conforms	6.4	169	4 h, 250 GB	6.4	0.0	0.0	merge	brain05, 2026-11-03
3	280	280	conforms	9.1	166	10 min, 250 GB	9.1	0.1	0.1	merge	brain01, 2026-11-03
4	2940	2940	conforms	21.9	3071	10 min, 250 GB	21.6	0.4	0.3	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	30	conforms	6.9	165	4 h, 250 GB	6.9	0.1	0.0	merge	brain05, 2026-11-03
3	280	280	conforms	5.4	182	10 min, 250 GB	5.3	0.2	0.1	merge	brain01, 2026-11-03
4	2940	2940	conforms	21.8	3022	10 min, 250 GB	21.6	0.3	0.1	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	2940	2822400	97	violation	26.3	420	48 h, 250 GB	10.0	26.3	0.0	merge	brain15, 2026-11-07

twins offline, Rust implementation against its Java twin: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	144	1	violation	7.4	166	10 min, 250 GB	7.1	0.1	0.3	merge	brain01, 2026-11-03
3	280	14400	12	violation	9.1	166	10 min, 250 GB	9.1	0.5	0.0	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **asyn-fault**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	36		conforms*	7.3	153	10 min, 250 GB	7.3	0.1	0.0	merge	brain01, 2026-11-03
3	280	400	2	violation*	5.5	182	10 min, 250 GB	5.3	0.1	0.2	merge	brain01, 2026-11-03
4	2940	4900	23	violation*	22.6	3022	10 min, 250 GB	21.6	0.3	1.0	merge	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). skipped: a smaller size had reached its limit: 4 bag asyn-fault.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 fifo asyn-fault, 5 fifo correct, 8 fifo asyn-fault, 8 fifo correct, 9 fifo correct, 9 fifo asyn-fault.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **asyn-fault**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	144	144	conforms	4.7	163	10 min, 250 GB	0.1	4.7	0.0	merge	brain01, 2026-11-03
3	14400	14400	conforms	25.3	4412	10 min, 250 GB	0.5	25.1	0.2	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **asyn-fault**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	36	36	conforms	4.1	160	10 min, 250 GB	0.1	4.1	0.1	merge	brain01, 2026-11-03
3	400	400	conforms	9.6	160	10 min, 250 GB	0.1	9.6	0.0	merge	brain01, 2026-11-03
4	4900	4900	conforms	14.9	3371	10 min, 250 GB	0.3	14.9	0.0	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	30	conforms	6.4	169	4 h, 250 GB	0.0	6.4	0.0	merge	brain05, 2026-11-03
3	280	280	conforms	9.2	166	10 min, 250 GB	0.1	9.1	0.1	merge	brain01, 2026-11-03
4	2940	2940	conforms	21.6	3071	10 min, 250 GB	0.4	21.6	0.1	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	30	conforms	6.9	165	4 h, 250 GB	0.1	6.9	0.0	merge	brain05, 2026-11-03
3	280	280	conforms	5.4	182	10 min, 250 GB	0.2	5.3	0.1	merge	brain01, 2026-11-03
4	2940	2940	conforms	21.9	3022	10 min, 250 GB	0.3	21.6	0.2	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	144	1	violation	4.8	163	10 min, 250 GB	0.0	4.7	0.1	merge	brain01, 2026-11-03
3	280	14400	12	violation	25.1	4412	10 min, 250 GB	0.1	25.1	0.0	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **asyn-fault**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	36		conforms*	4.1	160	10 min, 250 GB	0.1	4.1	0.0	merge	brain01, 2026-11-03
3	280	400	2	violation*	9.6	160	10 min, 250 GB	0.2	9.6	0.0	merge	brain01, 2026-11-03
4	2940	4900	23	violation*	15.1	3371	10 min, 250 GB	0.3	14.9	0.2	merge	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). skipped: a smaller size had reached its limit: 4 bag asyn-fault.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 fifo asyn-fault, 5 fifo correct, 8 fifo asyn-fault, 8 fifo correct, 9 fifo correct, 9 fifo asyn-fault.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **asyn-fault**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	144	144	conforms	5.3	174	10 min, 250 GB	5.3	0.1	0.0	merge	brain01, 2026-11-03
3	14400	14400	conforms	39.1	4571	10 min, 250 GB	39.0	0.5	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **asyn-fault**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	36	36	conforms	7.7	161	10 min, 250 GB	7.7	0.1	0.1	merge	brain01, 2026-11-03
3	400	400	conforms	6.7	184	10 min, 250 GB	6.4	0.1	0.2	merge	brain01, 2026-11-03
4	4900	4900	conforms	15.8	3401	10 min, 250 GB	15.4	0.3	0.4	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	30	conforms	7.0	160	4 h, 250 GB	7.0	0.0	0.0	merge	brain05, 2026-11-03
3	280	280	conforms	5.6	170	10 min, 250 GB	5.6	0.1	0.0	merge	brain01, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	2940	2940	conforms	17.0	2291	10 min, 250 GB	16.9	0.4	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	30	conforms	6.8	162	4 h, 250 GB	6.8	0.1	0.0	merge	brain05, 2026-11-03
3	280	280	conforms	5.0	176	10 min, 250 GB	4.9	0.2	0.1	merge	brain01, 2026-11-03
4	2940	2940	conforms	15.3		10 min, 250 GB	15.3	0.3	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair asyn-fault, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	2940	2822400	97	violation	26.3	2507	48 h, 250 GB	10.6	26.3	0.0	merge	brain15, 2026-11-07

context-free twins offline, Rust implementation against the context-free Java port: pair asyn-fault, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	144	1	violation	5.2	176	10 min, 250 GB	5.1	0.1	0.1	merge	brain01, 2026-11-03
3	280	14400	12	violation	5.7	169	10 min, 250 GB	5.6	0.5	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair asyn-fault, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	36		conforms*	4.4	165	10 min, 250 GB	4.3	0.1	0.0	merge	brain01, 2026-11-03
3	280	400	2	violation*	4.9	176	10 min, 250 GB	4.9	0.1	0.0	merge	brain01, 2026-11-03
4	2940	4900	23	violation*	15.5	2772	10 min, 250 GB	15.3	0.3	0.2	merge	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). skipped: a smaller size had reached its limit: 4 bag asyn-fault.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 fifo asyn-fault, 5 fifo correct, 8 fifo asyn-fault, 8 fifo correct, 9 fifo correct, 9 fifo asyn-fault.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair asyn-fault, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	144	144	conforms	5.3	174	10 min, 250 GB	0.1	5.3	0.0	merge	brain01, 2026-11-03
3	14400	14400	conforms	39.0	4571	10 min, 250 GB	0.5	39.0	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair asyn-fault, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	36	36	conforms	7.7	161	10 min, 250 GB	0.1	7.7	0.0	merge	brain01, 2026-11-03
3	400	400	conforms	6.4	184	10 min, 250 GB	0.1	6.4	0.0	merge	brain01, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	4900	4900	conforms	15.5	3401	10 min, 250 GB	0.3	15.4	0.1	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	30	conforms	7.0	160	4 h, 250 GB	0.0	7.0	0.0	merge	brain05, 2026-11-03
3	280	280	conforms	5.7	170	10 min, 250 GB	0.1	5.6	0.1	merge	brain01, 2026-11-03
4	2940	2940	conforms	17.0	2291	10 min, 250 GB	0.4	16.9	0.1	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	30	conforms	6.8	162	4 h, 250 GB	0.1	6.8	0.0	merge	brain05, 2026-11-03
3	280	280	conforms	5.3	176	10 min, 250 GB	0.2	4.9	0.4	merge	brain01, 2026-11-03
4	2940	2940	conforms	15.3	2772	10 min, 250 GB	0.3	15.3	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	144	1	violation	5.3	174	10 min, 250 GB	0.0	5.3	0.0	merge	brain01, 2026-11-03
3	280	14400	12	violation	39.1	4572	10 min, 250 GB	0.1	39.0	0.1	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **asyn-fault**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	30	36		conforms*	7.8	161	10 min, 250 GB	0.1	7.7	0.1	merge	brain01, 2026-11-03
3	280	400	2	violation*	6.4	184	10 min, 250 GB	0.2	6.4	0.0	merge	brain01, 2026-11-03
4	2940	4900	23	violation*	15.5	3401	10 min, 250 GB	0.3	15.4	0.1	merge	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 4 bag asyn-fault, 8 fifo asyn-fault.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 fifo asyn-fault, 5 fifo correct, 8 fifo correct, 9 fifo correct, 9 fifo asyn-fault.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **asyn-fault**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	144	144	conforms	6.5	172	1 h, 250 GB		brain01, 2026-11-03
3	14400	14400	conforms	35.4	4724	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **asyn-fault**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	36	36	conforms	6.4	165	1 h, 250 GB		brain01, 2026-11-03
3	400	400	conforms	8.9	176	1 h, 250 GB		brain01, 2026-11-03
4	4900	4900	conforms	23.0	4020	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	30	30	conforms	3.4	182	1 h, 250 GB		brain01, 2026-11-02
3	280	280	conforms	8.6	173	1 h, 250 GB		brain01, 2026-11-03
4	2940	2940	conforms	19.5	2674	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	30	30	conforms	3.4	156	1 h, 250 GB		brain01, 2026-11-02
3	280	280	conforms	8.8	170	1 h, 250 GB		brain01, 2026-11-03
4	2940	2940	conforms	20.4	2705	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **asyn-fault**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	30	144	12	violation	7.0	176	1 h, 250 GB	4119	brain01, 2026-11-03
3	280	14400	3438	violation	35.4	4575	1 h, 250 GB	4064	brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **asyn-fault**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	30	36		conforms*	6.4	156	1 h, 250 GB		brain01, 2026-11-03
3	280	400	32	violation*	9.1	183	1 h, 250 GB	4057	brain01, 2026-11-03
4	2940	4900	1218	violation*	23.2	4255	1 h, 250 GB	5178	brain01, 2026-11-03

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 8 fifo correct, 8 fifo asyn-fault, 9 fifo asyn-fault, 9 fifo correct.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **asyn-fault**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	144	144	conforms	3.6	157	1 h, 250 GB		brain01, 2026-11-02
3	14400	14400	conforms	26.7	5907	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **asyn-fault**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	36	36	conforms	3.4	164	1 h, 250 GB		brain01, 2026-11-02
3	400	400	conforms	4.5	203	1 h, 250 GB		brain01, 2026-11-02
4	4900	4900	conforms	15.6	4079	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair correct, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	30	30	conforms	3.2	172	1 h, 250 GB		brain01, 2026-11-02
3	280	280	conforms	4.2	204	1 h, 250 GB		brain01, 2026-11-02
4	2940	2940	conforms	13.9	3634	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	30	30	conforms	3.1	156	1 h, 250 GB		brain01, 2026-11-02
3	280	280	conforms	4.2	210	1 h, 250 GB		brain01, 2026-11-02
4	2940	2940	conforms	14.7	3647	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair asyn-fault, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	30	144	12	violation	3.2	172	1 h, 250 GB	2	brain01, 2026-11-02
3	280	14400	3438	violation	4.7	204	1 h, 250 GB	4	brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair asyn-fault, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	30	36		conforms*	3.1	156	1 h, 250 GB		brain01, 2026-11-02
3	280	400	32	violation*	4.3	210	1 h, 250 GB	3	brain01, 2026-11-02
4	2940	4900	1218	violation*	14.8	3647	1 h, 250 GB	3	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).
planned but never scheduled (grid points beyond the ladder): 8 fifo correct, 8 fifo asyn-fault, 9 fifo asyn-fault, 9 fifo correct.
skipped: a smaller size had reached its limit: 8 fifo asyn-fault.